Karan Singh

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SUMMARY

Cloud engineer with over 4 years experience, recognized for achieving annual cost savings exceeding $177,000 through strategi c optimizationinitiatives within 6 months. Skilled at crafting scalable cloud solutions tailored for diverse projects by utilizing multi-cloud platforms.

SKILLS

- **Cloud & Databases: AWS (EC2, VPC, S3, IAM, CloudFormation), Azure, GCP, DevOps, VMware, MySQL, MongoDB, PostgreSQL, NoSQL**
- **Languages: Python, C, C++, JavaScript, NodeJS, Bash/Shell scripting, HTML, CSS**
- **Frameworks & Tools: Docker, Kubernetes, Terraform, Jenkins, Ansible, GitLab CI/CD, REST API, Redis, Jira, Microservices**
- **Others: Git, Linux, Unix, Mac, Windows, Nginx, System Design, Networking, Operating Systems, VMWare, Machine Learning, AI**
- **Certifications: AWS Certified Cloud Practitioner, AWS Fundamentals Specialization, Google Cloud Big Data and Machine Learning Fundamentals**

EXPERIENCE

AWS Admin Intern

Cypress Creek Renewables **June 2023 - December 2023，Santa Monica, CA**

- Optimized cloud infrastructure by restructuring architecture, resolving performance bottlenecks, and managing resource allocation, reducingannual cloud costs by $177,062 (25.3%) from a $700,000 budget.
- Collaborated with cross-functional teams to automate infrastructure management using Infrastructure as Code (IaC), EKS, and ECS, resulting inimproved operational efficiency and system scalability.
- Gained hands-on experience in system administration and troubleshooting of Linux environments and cloud networking (e.g., TCP/IP,HTTP/s, DNS), resolving performance and connectivity issues using tools like traceroute, curl, ping, tcpdump, and Wireshark.

Associate Software Engineer

Accenture **June 2021 - July 2022，Bangalore, India**

- Migrated an event-driven architecture to AWS EventBridge, reducing message delay by 30% and improving integration with other AWS servicesto support a high-availability full-stack solution.
- Designed and implemented a scalable cloud architecture using Node.js, AWS Lambda, S3, and RDS, enhancing system performance andreducing latency across the application stack.
- Developed trigger-based events and REST APIs for a high-traffic web application, achieving a 40% increase in response rates and significantlyimproving the user experience.

Cloud Engineer

Advancing Medical Research, Innovation and Technology Welfare Society **May 2019 - May 2021，Jabalpur, India**

- Led full-stack development efforts in migrating on-premises infrastructure to AWS through microservices architecture using ECS, EKS &Lambda while optimizing performance and reducing operational costs by 30%.
- Created and deployed Jenkins pipeline for continuous integration and continuous delivery (CI/CD) of microservices, automating workflows andreducing deployment time by 50%.
- Enhanced system stability by resolving over 30 DNS issues, improving performance for cloud-based applications and reducing incident responsetime by 35%.

Cloud Engineer Intern

UnoDogs (2020 Google Android Developer Challenge winner) **May 2018 - April 2019，New Delhi, India**

- Developed cost-efficient full-stack architecture by optimizing AWS Reserved and Spot Instances, reducing cloud expenses by 38% whilemaintaining service quality.
- Conducted extensive evaluation of GCP, Azure, and AWS to choose the best cloud platform based on project requirements, ensuring costoptimization and performance improvement.
- Revamped GCP-based infrastructure for an iOS/Android app with 3,250 active users, reducing downtime by 70% and improving app performanceacross platforms.

EDUCATION

Master of Science in Computer Science

The University of Texas at Arlington • Arlington, Texas • 2024

Bachelor of Science in Computer Science

Dayananda Sagar University • Bangalore, India • 2021

PROJECTS

Earthquake Data Analysis and Visualization (Python, Flask, D3.js, Azure, Redis, SQL)

- Instituted a robust **CI/CD** pipeline integrating **Azure SQL** for queries, **Redis** for caching, and **Python** on Azure App Service, boosting web app performance by over 25%.
- Developed a data analysis website to visualize earthquake data, integrating Redis for data caching and Azure SQL for querying large datasets.
- Enhanced system performance by 25% through optimized caching strategies and cloud resource allocation.

Dispose-It (Python, AWS, R-CNN, SQL)

- Implemented an instance segmentation model using Mask R-CNN with an 87% accuracy rate to identify waste materials for an AI-driven waste management system.
- Developed a three-stage pipeline for detection, 3D reconstruction, and volume estimation, optimizing the overall accuracy of object identification in real-time environments.